



RMC.BestFit 2.0 Verification Report

Numerical verification, independent package parity,
published benchmarks, and scientific finding traceability

RISK MANAGEMENT CENTER

Verification program draft
RMC.Numerics 2.1.4 dependency
24 July 2026



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This report is generated from the living verification Markdown. A claim is verified only when its focused test, independent oracle, tolerance, and result are recorded.

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CHAPTER 01

RMC.BestFit 2.0 Verification Report

This is the living formal verification and validation record for `RMC.BestFit.Verification`. It separates scientific numerical evidence from fast unit and regression coverage.

Current status

Program area	Status	Evidence
Repository integration	Validated 24 July 2026	Debug Verification rebuild: 0 warnings/errors; Release solution log: 0 Verification project references
Public API baseline	Captured	Exact exported type/member/enum baseline enforced by <code>PublicApiCompatibilityTests</code>
Fast regression gate	Passed	Core 3,032; UI 564; App 427; 0 failed and 0 skipped
External environments	Locked	R 4.4.3 with 131 packages; Python with 15 packages
Test ownership audit	In progress	Initial duplicate removals and <code>FittingAnalysis</code> split are recorded in Test Inventory
Distribution fitting	Passed	All 15 family-specific and both multi-candidate external-oracle methods passed; TR-001, TR-009, and TR-063 are verified; TR-002 and TR-064 are rejected non-defects; TR-010 is fixed by regression. See Distribution Fitting
Model estimation and diagnostics	Planned after off-ramp	Model Estimation
Bulletin 17C and later phases	Planned	Scientific Review Findings

Evidence rule

A passing build, optimizer convergence, or reproduction of an internal result is not verification. A claim is verified only when its test has an analytical, independently implemented, external-package, or published-result oracle with declared provenance and tolerance.

Execution rule

The full Verification project is never run as one suite. Each result in this report is produced by exactly one fully qualified method through:

```
.\scripts\run-verification-test.ps1 -Test Namespace.Class.Method
```

The runner rejects broad filters and requires exactly one TRX result.

Traceability states

- **Planned:** no executable scientific test yet.
- **Ready - focused run:** the test compiles and may be executed only as one exact fully qualified method through the guarded runner.
- **Passed:** the focused TRX and oracle artifact have been reviewed.
- **Failed:** the focused test contradicted the claim or exposed a defect.
- **Blocked:** an external dependency, upstream correction, or scientific decision prevents completion.

Methodology | References

CHAPTER 02

Verification Methodology

Purpose

The program establishes claim-specific evidence for statistical correctness, numerical stability, and external parity. Unit tests remain in `RMC.BestFit.Tests`; this report covers only numerical verification and validation.

Evidence hierarchy

1. Closed-form analytical solution.
2. Independently implemented algorithm.
3. Trusted external package with a documented parameterization crosswalk.
4. Published worked example or table.
5. Simulation coverage with a predeclared design and Monte Carlo acceptance interval.

Regression pins and estimator smoke tests are useful engineering checks but are not independent verification.

Reproducibility contract

Each external artifact records the source dataset, package and runtime versions, generator path, command, seed, tolerance and rationale, SHA-256 hash, generation date, and affected finding IDs. R and Python generate committed CSV or JSON files. C# tests do not invoke those runtimes.

Tolerance policy

Comparison	Default
Deterministic analytical arithmetic	Absolute tolerance approximately 10^{-10}
Cross-language log likelihood and criteria	Absolute 10^{-8} , relative 10^{-7}
Optimizer parameter parity	Scaled 10^{-5} for comparable parameter-converged configurations; scaled 10^{-4} for objective-converged global versus parameter-converged local optimizers; tight objective parity is required in both cases
PSIS aggregate values	10^{-6}
Pareto k_i	10^{-4}
Coverage	Nominal rate must lie in the predeclared binomial interval

Exceptions must be documented before the C# result is observed.

Defect off-ramp

When independent evidence confirms a defect, scientific progression stops. The finding is updated and a focused correction plan is presented. Production correction begins only after approval and must preserve public signatures and serialization whenever possible.

After a correction, the three fast test projects, XML documentation gate, public API compatibility check, focused verification test, technical-reference chapter, finding entry, and this report must all agree before the finding is marked fixed.

CHAPTER 03

Verification Test Inventory

Ownership rule

Fast behavior, validation, serialization, property, state, event, exception, and regression tests belong in `RMC.BestFit.Tests`. Numerical verification remains only when a test uses an analytical, external, published, independently implemented, recovery, or coverage oracle.

Initial migration - 24 July 2026

Verification source	Disposition	Fast-test destination or coverage
<code>ModelEstimation/InfluenceDiagnosticsTests.cs</code>	Removed as unit/DTO coverage	Existing <code>Diagnostics/InfluenceDiagnosticsTests.cs</code>
<code>ModelEstimation/PointwiseLogLikelihoodTests.cs</code>	Removed as decomposition/unit coverage	Existing univariate, bivariate, time-series, mixture, point-process, and spatial unit tests
<code>ModelEstimation/PredictiveChecksTests.cs</code>	Removed as unit/regression coverage	Existing expanded prior/posterior predictive and result DTO unit tests
<code>DistributionFitting/FittingAnalysisTests.cs</code>	Split	Deterministic state/event/regression cases moved to <code>FittingAnalysisRegressionTests</code> ; published-data comparisons retained

Four legacy assertions were not retained: a probability-ordinate mutation test contradicted the established no-refit contract; generic large-sample and distribution-success counts had no oracle; and the outlier smoke test reproduced TR-010 (IsEstimated true with zero successful candidates). TR-010 is recorded as a confirmed defect, and its post-fix regression will be added with the approved correction.

Remaining audit

The following mixed files require method-level splitting before their domain phase begins:

Area	Mixed files
Model estimation	<code>GeneralizedMethodOfMomentsTests.cs</code> , <code>MaximumAPosterioriTests.cs</code>
Rating curve	<code>RatingCurveTests.cs</code> , <code>RatingCurveAnalysisTests.cs</code>
Time series	Model and analysis test files containing both constructors/properties and estimator recovery
Bivariate and spatial	Files containing both DTO/state checks and fitted-model recovery

The audit is intentionally marked **in progress**. No remaining mixed file is represented as verification-grade until its methods have been classified and moved.

CHAPTER 04

Distribution Fitting Verification

Scope

The first phase verifies the 15 supported univariate families, the fitting pipeline, goodness-of-fit metrics, and findings TR-001, TR-002, TR-009, TR-010, TR-063, and TR-064.

Distribution matrix

Family	Primary oracle	Secondary oracle	Status
Normal	Closed-form MLE	SciPy	Passed - SciPy parameter, likelihood, CDF, and quantile parity
Log10-Normal	Closed-form transformed MLE	SciPy	Passed - analytical and SciPy parity
Ln-Normal	Closed-form transformed MLE	SciPy	Passed - SciPy parity
Exponential	Closed form	SciPy	Passed - SciPy parity
Gamma	SciPy	Imomco	Passed - SciPy parity
GEV	SciPy	Imomco	Passed - SciPy parity
Generalized Logistic	Imomco	Published hydrology formulation	Passed - Imomco parity
Generalized Normal	Imomco	Published hydrology formulation	Passed - Imomco parity
Generalized Pareto	SciPy differential evolution	Imomco	Passed - global-optimum SciPy parity
Gumbel	SciPy	Imomco	Passed - SciPy parity
Kappa Four	Analytical branch identities	SciPy	Passed - zero-shape identities and SciPy finite-shape parity
Logistic	Closed form	SciPy	Passed - SciPy parity
Log-Pearson III	Transformed Pearson III	SciPy differential evolution	Passed - global-optimum SciPy parity
Pearson III	SciPy	Imomco	Passed - SciPy parity
Weibull	SciPy	Imomco	Passed - SciPy parity

TR-001 - Kappa Four zero primary shape

For $\kappa = 0$, $h \neq 0$, $z = (x - \xi)/\alpha$, the implemented CDF is

$$F(x) = [1 - h \exp(-z)]^{1/h}.$$

Independent differentiation and inversion give

$$f(x) = \frac{\exp(-z)}{\alpha} [1 - h \exp(-z)]^{1/h-1}$$

and

$$F^{-1}(p) = \xi - \alpha \log \left(\frac{1 - p^h}{h} \right).$$

The focused analytical tests are:

- `RMC.BestFit.Verification.DistributionFitting.KappaFourZeroShapeVerificationTests.ZeroShapeNonzeroHondo_PdfMatchesAnalyticalDerivative`
- `RMC.BestFit.Verification.DistributionFitting.KappaFourZeroShapeVerificationTests.ZeroShapeNonzeroHondo_QuantileInvertsCdf`

Disposition: Confirmed defect. Both exact methods failed against the baseline implementation.

Implementation: Fixed without public API changes in RMC.Numerics commit `3e058ebe5917817f3dde5e3b2ed6574d6bab083e`.

Verification: Passed. The PDF derivative and quantile/CDF round-trip methods were run separately through the guarded runner at absolute tolerance `1e-10`. The complete Numerics .NET 10 unit gate passed 1,905 tests with zero failures or skips. The evidence artifact records the baseline failures, passing results, source commits, and TRX paths.

TR-002 - finite Kappa shape pairs

The Kappa Four support changes with both finite shape parameters. Restrictions on the existence of particular moments do not imply that the distribution parameters themselves are invalid. The regression therefore spans all four sign combinations with (κ, h) equal to $(-0.25, -0.5)$, $(-0.25, 0.5)$, $(0.25, -0.5)$, and $(0.25, 0.5)$.

The focused unit method is:

- `Distributions.Univariate.Test_KappaFour.Test_K4_FiniteShapePairsHaveConsistentSupport`

Disposition: Rejected non-defect. Every finite pair is admitted when the scale is positive; the corresponding support is enforced by the CDF, density, and quantile functions.

Verification: Passed on Numerics .NET 10 at CDF/quantile absolute tolerance `1e-10`. The method verifies monotone finite quantiles at five probabilities, positive median density, and finite endpoint behavior. The complete Kappa Four class passed 12 tests with zero failures or skips. No BestFit Verification method was added because this is a fast parameter-domain regression owned by Numerics. See the evidence artifact.

TR-009 - parameter-adjusted RMSE

For n paired observations and modeled values with k fitted parameters, the implemented statistic is

$$\text{RMSE} = \left[\frac{1}{n-k} \sum_{i=1}^n (\hat{y}_i - y_i)^2 \right]^{1/2}, \quad 0 \leq k < n.$$

The deterministic fixture uses $y = [0, 0, 0, 0]$, $\hat{y} = [1, 2, 3, 4]$, and $k = 1$. Its independent hand calculation is

$$\sqrt{\frac{1^2 + 2^2 + 3^2 + 4^2}{4 - 1}} = \sqrt{10} = 3.1622776601683795.$$

The focused analytical method is:

- `RMC.BestFit.Verification.DistributionFitting.GoodnessOfFitRmseVerificationTests.ParameterAdjustedRmse_UsesAllResidualsAndIsPermutationInvariant`

Disposition: Confirmed defect. The baseline returned `2.160246899469287`, demonstrating that the final residual was omitted when $k = 1$.

Implementation: Fixed without public API changes in RMC.Numerics commit `24bf9f98139b23400bf008df413b0d97330ccfd3`. All n residuals enter the numerator, $n - k$ is used only as the denominator, and invalid k values are rejected.

Verification: Passed at absolute tolerance `1e-12`, including paired-permutation invariance. The complete Numerics .NET 10 unit gate passed 1,907 tests with zero failures or skips. The evidence artifact records the baseline failure, corrected result, source commits, and TRX paths.

TR-010 - all-candidate failure reports overall success

A deterministic outlier fixture completed with `IsEstimated == true` while all 15 `FittedDistribution.FitSucceeded` values were false. Source inspection confirms that the outer run sets `IsEstimated = true` whenever the parallel loop itself completes, without requiring a successful candidate.

Disposition: Confirmed defect by deterministic behavioral reproduction and source inspection.

Implementation: Fixed without public API changes. `IsEstimated` and `AnalysisCompleted.Succeeded` now require at least one successful candidate; partial-success runs remain successful.

Verification: Passed by deterministic all-failed and partial-success regressions. The complete .NET 10 core gate passed 3,030 tests with zero failures or skips. No Verification-project method was added because this is a state-semantic regression, not a numerical validation claim. See the evidence artifact.

TR-063 - whole-series replacement refresh

For an uncensored exact sample with n observations and default Weibull plotting parameter, ascending order statistic i has nonexceedance probability

$$p_i = \frac{i}{n+1}.$$

The baseline whole-series setter left every new `ExactData.PlottingPosition` at zero. `FittingAnalysis` therefore consumed complemented probability one, evaluated infinite candidate quantiles, and assigned infinite RMSE even when MLE and information criteria were finite.

Disposition: Confirmed defect.

Implementation: Fixed without public API or XML-schema changes. A valid programmatic series replacement performs one derived-state refresh after attaching the new handlers. Invalid transient frames remain assignable and defer refresh. The XML constructor suppresses all four replacement refreshes, preserves serialized positions exactly, and reprocesses effective threshold counts once.

The focused analytical method is:

- `RMC.BestFit.Verification.DistributionFitting.FittingAnalysisCriteriaVerificationTests.ExactSeriesReplacement_ProducesAnalyticalWeibullPositions`

Verification: Passed for all 39 oracle observations at absolute tolerance `1e-12`. Fast regressions also prove that deliberately non-derived serialized positions survive a round trip and that non-finite transient input retains the previous non-throwing behavior. The complete Debug regression gate passed Core 3,032, UI 564, and App 427 tests with zero failures or skips; the public API baseline and enforced XML-documentation build also passed. See the evidence artifact.

TR-064 - distribution-fitting optimizer tolerance

The common-data external validation fits Gumbel, Normal, and Logistic to one deterministic sample. BestFit uses differential evolution, whose stopping rule detects convergence in objective values across the population. The SciPy oracle uses a local configuration that converges in parameter or gradient space. Minor coordinate differences are therefore expected even when the fitted likelihoods are effectively identical.

Gumbel parameter	SciPy oracle	BestFit	Relative error	<code>1e-12</code> scaled acceptance
Gumbel location	93.11234799935337	93.1129321294911	6.27e-6	Passed
Gumbel scale	13.157628567998076	13.157823249599968	1.4796e-5	Passed

The exact focused methods are:

- `RMC.BestFit.Verification.DistributionFitting.FittingAnalysisCriteriaVerificationTests.FittedParameters_MatchIndependentOracle`
- `RMC.BestFit.Verification.DistributionFitting.FittingAnalysisCriteriaVerificationTests.CriteriaRankingWeightsAndConfiguredOrder_MatchIndependentOracle`

Disposition: Rejected non-defect. No production change was made.

Tolerance rationale: Parameters use `1e-4` scaled tolerance for this global-versus-local optimizer comparison. Maximum log likelihood, AIC, and BIC retain the tighter cross-language tolerance of `1e-8` absolute plus `1e-7` relative. RMSE magnitudes are not compared at different optimizer-returned parameter vectors. Instead, the BestFit RMSE equation is checked directly to `1e-10`, inverse-RMSE weights are checked from the actual RMSE values to `1e-12`, and the cross-optimizer RMSE ranking must agree exactly.

Verification: Both focused methods passed. The parameter run is recorded at `TestResults/VerificationFocused/20260725-081305-...`; the criteria, ranking, and weight run is recorded at `TestResults/VerificationFocused/20260725-081522-...`. See the evidence artifact.

Log10-Normal analytical verification

For $x_i = \log_{10}(y_i)$,

$$\hat{\mu} = \frac{1}{n} \sum_i x_i, \quad \hat{\sigma}^2 = \frac{1}{n} \sum_i (x_i - \hat{\mu})^2.$$

For the deterministic log-space values [1.1, 1.4, 1.7, 2.0, 2.3, 2.6, 2.9], the closed-form estimates are $\hat{\mu} = 2$ and $\hat{\sigma} = 0.6$, where the variance divisor is n . On the original measurement scale, the log likelihood includes the transformation Jacobian:

$$\ell(\mu, \sigma) = -n \log(\sigma \sqrt{2\pi}) - \frac{1}{2\sigma^2} \sum_i (x_i - \mu)^2 - \sum_i \log(y_i \ln 10).$$

The closed-form log likelihood is `-44.431208784723104`; the median is $(10^2=100)$, and the 0.9 quantile is `587.3959385303014`.

The focused method is:

- `RMC.BestFit.Verification.DistributionFitting.Log10NormalFittingVerificationTests.ClosedFormMle_LikelihoodCdfAndQuantileMatchAnalyticalOracle`

Verification: Passed. BFGS used verification-only absolute and relative convergence tolerances of `1e-12`. Parameters passed at absolute tolerance `1e-5`, maximum log likelihood at `1e-8`, the direct and pointwise analytical likelihood at `1e-10`, the median CDF at `1e-12`, and the analytical quantile at `1e-9`. The fitted quantile uses the declared optimizer-scale relative tolerance. See the evidence artifact.

External family-oracle execution

Thirteen exact methods in `ScipyDistributionFittingVerificationTests` and two exact methods in `LmomcoDistributionFittingVerificationTests` were run separately through the guarded runner. Each method verifies the fitted parameter vector, maximized data log likelihood, representative CDF values, and representative quantiles. The C# tests consume only committed JSON and never invoke Python or R at runtime.

SciPy 1.16.1 supplies the overlapping family implementations. Generalized Pareto and Log-Pearson III use deterministic differential evolution in the generator because SciPy's default local start converged to an inferior stationary point for the declared fixtures. R 4.4.3 with `lmomco` 2.5.7 supplies Generalized Logistic and Generalized Normal. Parameter conversions, package versions, seeds, generator commands, and source hashes are recorded in:

- SciPy family oracles
- Imomco family oracles
- Verification data manifest

All 15 family-specific methods and both multi-candidate `FittingAnalysis` methods passed. The result verifies the individual family implementations and the declared common-data ranking, criteria, RMSE-formula, and weighting claims.

CHAPTER 05

Model Estimation and Diagnostics Verification

Status

Planned after the TR-001 off-ramp is resolved.

Log10-Normal prior and penalty experiment

A deterministic symmetric log10 dataset and an outlier variant will compare MLE, flat-prior MAP, and Bulletin 17C GMM. A Gaussian prior on μ is mapped to the quadratic penalty using the implementation convention $MSE = \tau_\mu^2/n$, subject to the TR-033 objective-scale correction.

Regime	Prior scale and center	Expected location effect	Expected variance effect
Wide centered	$100SE_L$, centered at MLE	None	Negligible
Narrow centered	$0.1SE_L$, centered at MLE	None	Large contraction
Narrow shifted	$0.1SE_L$, shifted by $2SE_L$	Material shift	Large contraction

Exact leave-one-out refits provide the observation-influence oracle. Full-Hessian and deletion covariance comparisons provide the leverage oracle; no sum-to- p assertion is accepted without a derivation.

External model-comparison oracles

- R `loo` 2.10.0 for WAIC and PSIS-LOO.
- R `BayesianTools` 0.1.9 for the selected DIC convention.
- R `gmm` 1.9-1 for estimates, covariance, and Hansen's J .
- R `bbmle` 1.0.25.1 for nuisance-reoptimized profiles.
- Python `ArviZ` 1.2.0 as a secondary WAIC/LOO comparison.

The fixtures will use committed deterministic posterior draws and pointwise log-likelihood matrices so package parity is not confounded with sampler randomness.

CHAPTER 06

Verification References

- Asquith, W. H. `lmomco`: L-moments and hydrologic distributions, CRAN package 2.5.7.
- Bolker, B. and R Development Core Team. `bbmle`: tools for general maximum likelihood estimation, CRAN package 1.0.25.1.
- Chausse, P. `gmm`: generalized method of moments and generalized empirical likelihood, CRAN package 1.9-1.
- Hartig, F. et al. `BayesianTools`: general-purpose MCMC, SMC, and Bayesian diagnostics, CRAN package 0.1.9.
- Vehtari, A. et al. `loo`: efficient leave-one-out cross-validation and WAIC, CRAN package 2.10.0.
- SciPy community. `scipy.stats` reference, version 1.17.1.
- RMC-BestFit version 1 verification report, [docs/reports/RMC-TR-2020-02 - Verification of the Bayesian Estimation and Fitting Software.pdf](#).
- RMC.BestFit 2.0 Technical Reference and Scientific Review Findings.